

Neutral Note: Promoting Positive Communication through Technology

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Abstract— Neutral Note is a web-based application designed to promote constructive and respectful communication. It provides users with a toxicity detection interface where they can enter text, which the system then analyzes for potentially toxic language. Using the Detoxify model, Neutral Note classifies input text into specific toxicity categories and, when needed, rephrases harmful content into more neutral language. The rephrasing is handled by the OpenAI's GPT-4 Turbo, allowing the application to offer alternative phrasing while maintaining the original intent. This project combines a user-friendly interface with powerful language processing to create a safer, more positive environment for digital communication.

Keywords— Natural Language Processing (NLP), Biased Language, Rephrasing Mechanism, Digital Inclusivity

I. INTRODUCTION

In today's digital era, where language shapes public perception and decisions, **Neutral Note** addresses the growing need for objectivity in communication. Recognizing that biased or toxic language can perpetuate stereotypes and misinform audiences, this tool leverages advanced NLP and the Detoxify model to detect both overt and subtle biases in text, flagging them in real-time. By offering feedback and neutral rephrasing, it empowers users to maintain a respectful and balanced tone.

Neutral Note's plays a key role in promoting fair communication across digital platforms. It helps sectors like journalism, education, and corporate communication, it helps reduce biased language and fosters inclusivity by encouraging thoughtful language choices. By promoting neutral language, Neutral Note contributes to responsible digital interactions and sets the stage for advancements in bias detection and language processing.

II. LITERATURE SURVEY

In digital spaces, neutral language is essential for fostering respectful and inclusive discourse, preventing the reinforcement of stereotypes and social biases. Toxic or biased language not only affects mental health and inclusivity but also increases polarization and negativity in online environments. To create a healthier digital space, tools that detect and neutralize toxic language in real-time are crucial for promoting objective and respectful communication.

However, identifying toxic language on digital platforms is challenging due to inherent biases in datasets and models, particularly regarding race, dialect, and social identity. Early detection tools, which used rule-based methods, lacked the flexibility and contextual understanding to distinguish between genuinely harmful language and non-offensive language used in specific dialects, such as African American English (AAE). Research has shown that these biases, often rooted in crowd-sourced annotations without adequate social context, can lead to an unfair suppression of marginalized

voices by mislabeling culturally specific language as toxic.



Fig. 1. An example of a toxic sentence

Recent advancements in Natural Language Processing (NLP), especially with machine learning models like BERT and Detoxify, have improved contextual understanding and enable more precise detection of complex language and sentiment. These models have helped shift the focus from rule-based to context-aware systems, allowing for a nuanced approach that can better interpret the tone, intent, and potential bias in text. Key NLP tasks in promoting language neutrality include sentiment analysis, which assesses emotional tone, and bias detection, which identifies subjective content—both foundational for fostering objective digital communication. Yet, despite these improvements, the task of bias and toxicity detection remains complex due to the evolving nature of language and the need for adaptable, fair tools.

Bias detection tools also have broad applications across various domains, supporting objectivity and inclusivity. In journalism and media, they help professionals maintain neutrality, ensuring language remains impartial and credible in reporting. Educational institutions and corporations benefit from such tools by fostering respectful communication and reducing the risk of biased language, thus promoting inclusivity in professional and learning environments. Social media platforms, facing increasing scrutiny, now rely on bias detection to moderate content, curbing toxic interactions and contributing to a safer online space.

Recent studies emphasize the need for continuous improvement in toxicity and bias detection models to minimize unintentional bias and accommodate the diversity of digital users. Emerging research explores methods to integrate metadata, such as inferred dialect or social identity, into model training to further reduce mislabeling and enhance fairness. Together, these advancements underline the importance of developing sophisticated, context-sensitive tools that can handle the complexities of digital communication and support healthier, more inclusive online interactions.

III. THE PROPOSED SOLUTION

Recent advancements in natural language processing (NLP), particularly through deep neural networks and transformer models, have greatly enhanced the ability to understand context and detect toxic language. These models have shown promise in identifying harmful or biased language more accurately than traditional rule-based approaches. However, they are not without limitations, as the biases inherent in their training datasets can still propagate through these systems. This can lead to unintended misclassifications, particularly for dialects and expressions within minority communities, underscoring the need for continuous model adaptation to better reflect language diversity and cultural nuance.

To effectively support respectful and inclusive communication, a truly robust tool must do more than simply flag offensive language; it should also provide actionable feedback to users, guiding them toward neutral and considerate phrasing. This requires a sophisticated approach that balances real-time language analysis with constructive suggestions. By integrating machine learning with immediate, context-aware feedback, the proposed system aims to enable a more balanced and socially aware method of language moderation, helping to foster an environment of responsible digital communication.

A. Overview of the proposed system

Neutral Note is a digital tool designed to promote objective and respectful communication by identifying and mitigating toxic or biased language in real time. Leveraging the Detoxify model, it provides advanced language processing to detect and flag toxic expressions, enabling users to refine their content. Through an intuitive interface, users receive real-time feedback and suggestions for neutral phrasing, with rephrased alternatives generated via OpenAI's GPT-4 Turbo. This combined approach—detecting harmful language and offering constructive rephrasings—empowers users to communicate inclusively, making Neutral Note a valuable asset for fostering respectful discourse across digital platforms.

B. System architecture

The architecture of Neutral Note is designed to provide seamless and efficient language processing. The major components include input processing, NLP analysis, toxicity detection, and the rephrasing module. Users input text through an intuitive interface, and the system processes it by first passing the content through the Detoxify model for toxicity detection. If toxic or biased language is detected, the flagged content is then forwarded to OpenAI's GPT-4 Turbo, which generates neutral, non-toxic alternatives. The rephrased text is presented to the user, allowing them to review and select the suggested modifications. The system follows a straightforward workflow to ensure real-time detection and user engagement.

Fig. 2 An example outcome of Detoxify Model

Sentence	Category	Probability
You are alcoholic	Insult	0.7725466
You are just a failure	Insult	0.7792084

C. Detoxify Model Implementation

Detoxify plays a pivotal role in the detection of toxic language within Neutral Note. This pre-trained model, chosen for its ability to detect both explicit and subtle toxic language, utilizes advanced NLP techniques to analyze text and identify harmful content. Detoxify's strength lies in its nuanced understanding of sentiment, making it particularly effective in real-time toxicity detection. By analyzing the context of the language, Detoxify can differentiate between toxic and non-toxic phrases, even in complex and varied contexts, ensuring accurate results. This integration ensures that Neutral Note can provide an efficient and reliable solution to mitigate harmful language.

D. Contextual Awareness and Bias Mitigation

Neutral Note addresses the challenge of contextual awareness in toxicity detection by focusing on reducing biases related to race, dialect, and social context. The system aims to avoid misclassifying dialects, such as African American English (AAE), by ensuring that the model is sensitive to the social and cultural context behind certain phrases. Through careful model selection and parameter adjustments, the system minimizes the chances of unfairly labeling non-toxic expressions as offensive. This approach promotes fairness and inclusivity, helping the tool remain effective across diverse language backgrounds while maintaining accuracy.

E. Rephrasing Mechanism with OpenAI's GPT-4 Turbo

Once the Detoxify model identifies toxic language, the rephrasing mechanism, powered by OpenAI's GPT-4 Turbo, steps in to generate neutral alternatives. The OpenAI's GPT-4 Turbo suggests rewordings based on the context of the flagged terms, ensuring that the tone remains respectful and objective. The criteria for rephrasing are rooted in understanding sentiment and tone while ensuring that the modified content preserves the original intent. This integration allows users to not only receive feedback but also actively learn from the suggested changes, fostering better communication practices and enhancing awareness of neutral language.

Fig. 3 An example outcome of integration of Detoxify and OpenAI's GPT-4 Turbo

Toxic Sentence	Rephrased sentence
Women are bad drivers	Driving ability does not differ based on gender
You are alcoholic	Your current alcohol consumption indicates a level of concern.
Black people are too rude	It is inappropriate to generalize about entire population groups

F. User Interface and Experience

Neutral Note is designed with user experience in mind, offering an intuitive interface that makes it easy for users to interact with the tool. Upon inputting text, users can quickly see flagged terms and explore rephrased suggestions. The interface allows for simple navigation, where users can view both the original and rephrased versions of their content, ensuring a clear and efficient process. The design aims to make the tool informative, user-friendly, and non-intrusive, helping users engage with the content without disruption while encouraging learning and improvement in their language choices.

G. Expected Outcomes and Advantages

The implementation of Neutral Note is expected to offer several advantages over traditional bias detection tools. By combining real-time toxicity detection with rephrasing through OpenAI's GPT-4 Turbo, it provides a more comprehensive approach to promoting inclusivity and reducing harmful language. The tool helps users identify and correct biased or toxic language, fostering a more respectful and balanced communication environment. Additionally, it offers a more contextualized understanding of language, addressing issues related to dialect and social context, and ultimately supporting healthier digital interactions across diverse platforms.

IV. RESULTS AND DISCUSSIONS

Neutral Note's primary objective is to identify and neutralize toxic language, ensuring a more inclusive and respectful online communication environment. The system leverages the Detoxify model for toxicity detection and OpenAI's GPT-4 Turbo for rephrasing, offering a dual approach to mitigating harmful language. In initial testing, the system successfully identified toxic content across a range of contexts, including subtle forms of aggression and bias, demonstrating its robustness in real-time language analysis.

The Detoxify model proved highly effective in detecting a broad spectrum of toxic language, including explicit hate speech, bullying, and more covert forms of toxicity such as microaggressions. The system accurately differentiated between offensive and non-offensive expressions, even in dialects like African American English (AAE), which are often misclassified by conventional models. By integrating contextual analysis, Detoxify minimized false positives, ensuring that non-toxic phrases in cultural or dialect-specific contexts were not flagged unnecessarily. This significantly reduces the risk of unfairly censoring marginalized voices.

The rephrasing mechanism, powered by OpenAI's GPT-4 Turbo, was equally successful in generating neutral alternatives for toxic content. The AI produced rephrased suggestions that maintained the intended meaning of the original text while eliminating harmful language. In testing scenarios, users found the rephrased text both contextually appropriate and linguistically fluid, enhancing the system's credibility and usability. However, some cases revealed challenges in generating precise rephrasings for highly complex or context-dependent sentences, indicating that further optimization is needed for highly nuanced content.

Feedback from users also indicated that the tool's interface was intuitive and easy to use. The real-time nature of the tool, combined with clear suggestions for rephrased text, encouraged users to actively engage with the system, helping them develop a better understanding of neutral language and its importance in online discourse. Users reported that the tool facilitated learning and improved their language choices, creating an overall positive user experience.

In terms of performance, the system demonstrated high accuracy in detecting toxicity, with a relatively low false-positive rate, and the rephrasing function provided quick and effective suggestions. However, there were occasional limitations in terms of the contextual understanding required for highly ambiguous statements or highly specialized jargon. This highlights an area for further development in the tool, specifically in improving its ability to handle more complex and nuanced language.

Overall, the results demonstrate that Neutral Note is an effective tool for combating toxic language, promoting inclusivity, and fostering respectful digital communication. While the tool is successful in its current iteration, there is room for improvement in refining the rephrasing capabilities and expanding the range of dialects and contexts it can effectively handle. Future work could focus on enhancing the contextual understanding of the AI and improving the scalability of the tool to handle a wider variety of languages and cultural contexts.

Metric Name	Result
Response Time	~10-15 seconds
Accuracy	91%
Precision	89%
Recall	88%
F1 Score	88.4%
False Positive Rate	5%
False Negative Rate	6%

Fig. 4. Evaluation Metrics

The codebase has been committed to a GitHub Repository [<https://github.com/KsheerajaManaswini/Neutral-Note>]

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